# SHEAFIFICATION DOES NOT CHANGE STALKS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sheaves.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- There is a canonical map Pi(F)_x -> F_x left inverse to F_x -> Pi(F)_x, as described in the pointwise-product-stalk example. invoked at proof lines 1790-1793. Provides injectivity of F_x -> F#_x.
- Sections of F# are families satisfying property (*): locally they are represented by genuine sections of F. invoked at proof lines 1796-1801. Provides the local representative used for surjectivity.

*Source reference.* Original remark 007D: sources/stacks-sheaves/pinned-original-source.tex, source lines 1250–1250. Cited at proof lines 1790-1793 for the left inverse establishing injectivity; load-bearing inside the counted span.

**Lemma 0.2.** *Let $X$ be a topological space. Let $\mathcal{F}$ be a presheaf of sets on $X$. Let $x \in X$. Then $\mathcal{F}_x = \mathcal{F}^\#_x$.*

*Proof.* The map $\mathcal{F}_x \to \mathcal{F}^\#_x$ is injective, since already the map $\mathcal{F}_x \to \Pi(\mathcal{F})_x$ is injective. Namely, there is a canonical map $\Pi(\mathcal{F})_x \to \mathcal{F}_x$ which is a left inverse to the map $\mathcal{F}_x \to \Pi(\mathcal{F})_x$, see Example 007D. To show that it is surjective, suppose that $\overline{s} \in \mathcal{F}^\#_x$. We can find an open neighbourhood $U$ of $x$ such that $\overline{s}$ is the equivalence class of $(U, s)$ with $s \in \mathcal{F}^\#(U)$. By definition, this means there exists an open neighbourhood $V \subset U$ of $x$ and a section $\sigma \in \mathcal{F}(V)$ such that $s|_V$ is the image of $\sigma$ in $\Pi(\mathcal{F})(V)$. Clearly the class of $(V, \sigma)$ defines an element of $\mathcal{F}_x$ mapping to $\overline{s}$. □

*Date*: stacks-project:sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.